Semantic Derivation in AI-Generated Consciousness Theories: Benchmark Evidence from a Multi-Model Theory-Generation Experiment

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Abstract

Can large language models propose consciousness theories that introduce genuinely new ontological commitments, or do their outputs remain bounded within known theoretical families? We investigate this question through a systematic multi-model experiment in which seven state-of-the-art language models were prompted to formulate theories of consciousness based on multimodal sensory activation patterns. Generated theories ($\boldsymbol{N = 168}$; $\boldsymbol{n \approx 24}$ per model) were evaluated against a reference corpus of 46 known consciousness theory families using semantic similarity computed with SentenceTransformer embeddings, a three-stage computational-functionalism detection algorithm, and a manifold-based novelty diagnostic. Across all three evaluation axes, results converge on a single finding: 91.7% of theories ($\boldsymbol{n = 154}$) classified as hybrid recombinations of known frameworks, 7.1% ($\boldsymbol{n = 12}$) as direct literature instantiations, and 1.2% ($\boldsymbol{n = 2}$) as purely computational-functionalist; zero theories achieved confirmed framework-external novelty at any calibrated threshold. Threshold sensitivity analysis over the range $\mathbf{[0.20, 0.50]}$ consistently returned a gated novel rate of $\mathbf{0.00}$. Cross-model uniformity was striking: one-way ANOVA on the primary novelty metric yielded $\boldsymbol{F(6, 161) = 1.12}$, $\boldsymbol{p = 0.355}$, $\boldsymbol{\eta^2 = 0.040}$, the lowest effect size across the full four-test benchmark. The findings distinguish lexical and structural creativity—which models demonstrate in abundance—from ontological-semantic novelty, which is uniformly absent. We provide a reproducible protocol for auditing theory-level semantic novelty and discuss implications for the use of AI in scientific theory construction.

Keywords: consciousness theories, large language models, semantic novelty, computational creativity, theory generation, ontological derivation, embedding similarity

1 Introduction

Theory generation sits at the apex of scientific cognition. A researcher who can propose an entirely new theoretical framework does not merely extend existing knowledge—they reorganise the conceptual space within which further inquiry proceeds [1]. Whether artificial intelligence can perform this operation is therefore a question with deep implications for how we should deploy AI systems in knowledge-intensive fields [2, 3].

Large language models (LLMs) have achieved remarkable performance on a wide range of language tasks [4–6], including generating plausible-sounding text about scientific topics [7–9]. Yet there is a systematic gap between producing linguistically fluent, topically coherent output and actually proposing a theory that adds new ontological content to a domain [10, 11]. Fluency is necessary but not sufficient; the more demanding criterion is whether generated proposals can be traced back to recombinations of training-data primitives or whether they transcend the conceptual manifold the model was trained on.

Consciousness studies provides an ideal test domain for this question. The field has a well-established taxonomy of competing theoretical families—from Integrated Information Theory [12] and Global Workspace Theory [13, 14] to predictive-coding accounts [15] and quantum-substrate proposals [16]—making it possible to construct a comprehensive reference corpus against which novel outputs can be evaluated. The domain is also sufficiently sophisticated that surface-level fluency does not imply genuine theoretical innovation, and it is one where truly novel ontological commitments (e.g. introducing a non-computational substrate or abandoning multiple-realisability) would be immediately identifiable as framework-transcending.

This article presents a focused analysis of a consciousness theory-generation benchmark (Test 3 from a broader four-test AI creativity study). Seven state-of-the-art models—GPT-5.2, Claude-3.7-Sonnet, Gemini-3.1-Pro-Preview, DeepSeek-v3.2, Mistral-Large, LLaMA-3.3-70B-Instruct, and Perplexity-Sonar-Pro—each generated approximately 24 theories in response to a standardised prompt presenting multimodal sensory activation data. All 168 theories were then evaluated with three independent analysis pipelines: a computational-functionalism (CF) detection algorithm, a semantic-similarity traceability comparison against 46 known theory families, and a manifold-based novelty diagnostic using k-nearest-neighbour boundary estimation.

The main contributions of this article are:

1. A reproducible multi-axis evaluation protocol for measuring semantic novelty in AI-generated theoretical texts, applicable beyond the consciousness domain.
2. Quantitative evidence that theory-level output from seven diverse models is universally constrained to recombinations of known conceptual families, with zero confirmed instances of framework-external novelty.
3. Characterisation of the most recurrent theoretical families in AI consciousness output, revealing a biased repertoire dominated by synchrony-binding, adaptive-resonance, and dynamic-core traditions.
4. A threshold sensitivity analysis showing that the null result is robust to calibration choices across the full plausible parameter range.

The remainder of the article is organised as follows. Section 2 reviews the relevant background. Section 3 describes experimental design and evaluation protocol. Section 4 reports results. Section 5 interprets the findings. Section 6 acknowledges limitations, and Section 7 concludes.

2 Background and Related Work

2.1 Creativity Taxonomies and Theory Generation

Boden's influential taxonomy distinguishes three forms of creativity: *combinational* (novel combinations of familiar ideas), *exploratory* (novel ideas produced by exploring a given conceptual space), and *transformational* (ideas that require changing the structure of the space itself) [10]. Computational systems have demonstrated convincing combinational and exploratory creativity across many domains [17–20], but evidence for transformational creativity—where the output introduces a concept that cannot be derived from the prior conceptual basis—has remained elusive [11, 21].

Theory generation, as studied here, calls for the most demanding variant of transformational creativity: producing an ontological claim about the nature of consciousness that is not reducible to existing theoretical family memberships. This is distinct from proposing a new terminology, combining known mechanisms in an unusual pattern, or emphasising a neglected aspect of an existing framework. We refer to the capacity to introduce genuinely new ontological primitives—kinds of entities or processes not logically derivable from the training corpus—as *ontological creativity*, following the conceptual apparatus developed in related work on AI creativity limits [10].

2.2 The Consciousness Theory Landscape

Philosophical and neuroscientific theories of consciousness span a broad but finite space. The major families relevant to this study include:

- **Integrated Information Theory (IIT):** Consciousness correlates with integrated information (Φ); physical substrate is secondary to causal integration [12].
- **Global Workspace Theory (GWT):** Consciousness arises when information is broadcast from a central workspace to distributed modules; emphasises access and reportability [13, 14].
- **Predictive Processing / Free Energy:** The brain minimises prediction error by constructing generative models of sensory causes; consciousness reflects the hierarchical generative model state [15].
- **Attention Schema Theory (AST):** The brain constructs a model of its own attention, and this model is what we call consciousness [22].
- **Local Recurrent Processing:** Conscious experience requires recurrent local processing in visual cortex, not feedforward signals alone [23].
- **Synchrony and Binding:** Conscious integration is achieved through synchronous gamma-band oscillations binding distributed cortical representations [24].
- **Quantum Consciousness Theories:** Consciousness involves quantum-coherent processes in neuronal microtubules or other structures [16, 25].

- **Computational Functionalism:** Mental states are multiply realisable functional states; substrate is irrelevant so long as the functional organisation is preserved [26].

This taxonomy is not exhaustive but captures the families most densely represented in AI training corpora, and therefore provides the strongest possible test for detecting framework-external novelty: if models cannot escape even this well-documented repertoire, the constraint is robust.

2.3 Semantic Novelty and Traceability in AI Text

Recent work on evaluating AI-generated text has increasingly distinguished *surface novelty*—originality at the level of n-gram statistics or lexical choice—from *semantic novelty*—originality at the level of meaning and conceptual content [7, 8, 27, 28]. LLMs routinely achieve high surface novelty, generating text that does not literally repeat training examples, while producing outputs that map onto known semantic structures when measured via dense embeddings. Sentence-level transformer embeddings [29] provide a natural operationalisation of semantic similarity that is robust to paraphrasing and terminological variation.

Prior work on scientific text generation has shown that LLM outputs remain confined to the distributional properties of training data [4, 30]. Mechanistic interpretability studies confirm that generation proceeds via attention-mediated interpolation among training representations [31], providing a principled reason to expect systematic traceability. The formal counterpart—that gradient-based learning in continuous spaces cannot expand the intrinsic dimensionality of the representation manifold beyond the training data span—places the empirical expectation on firm mathematical ground.

2.4 The Representational Constraint Hypothesis

A key theoretical prediction motivating this study is the following: because LLMs learn continuous transformations of embeddings via gradient descent, their output manifold is necessarily a continuous deformation of the training data manifold [32]. Operationally, this implies that any generated text, however terminologically novel, should be locatable within the convex hull of training- derived embeddings. We term this the *representational constraint hypothesis* and design the manifold-based novelty diagnostic specifically to test it.

3 Materials and Methods

3.1 Experimental Task

Models were prompted with structured multimodal sensory activation data representing neural response patterns during three phenomenal states: alert wakefulness, resting state, and dreaming. The activation profiles encoded responses across six sensory modalities: visual, auditory, somatosensory, proprioceptive, interoceptive, and vestibular. The prompt instructed each model to propose a comprehensive theory explaining how these activation patterns relate to conscious experience, addressing three sub-questions: What is consciousness? How does consciousness relate to sensory processing? Why do these specific patterns correspond to different experiential states?

The experimental protocol required each model to generate a named theory with an identifiable acronym, a set of core ontological commitments (claims about the fundamental nature of consciousness), and predictions distinguishing the proposed theory from alternatives. Presenting structured quantitative data alongside an open-ended explanatory task ensured the prompt invited genuine theoretical innovation rather than eliciting recall of named theories from literature.

3.2 Models

Seven models participated in the experiment: GPT-5.2 [5], Claude-3.7-Sonnet, Gemini-3.1-Pro-Preview [6], DeepSeek-v3.2 [33], Mistral-Large [34], LLaMA-3.3-70B-Instruct [35], and Perplexity-Sonar-Pro. The ensemble spans transformer-based architectures with different training objectives (supervised pre-training plus RLHF alignment), mixture-of-experts variants, and parameter scales ranging from approximately 70 billion to several hundred billion parameters. Sampling temperature was fixed at $T = 0.7$ across all models to balance diversity with coherence. Each model generated $n \approx 24$ theories, yielding $N = 168$ theories in total.

3.3 Reference Corpus and Theory Family Taxonomy

A reference corpus of 46 known consciousness theory families was compiled from philosophical and neuroscientific literature spanning classical functionalism through quantum-substrate approaches. Each family was represented by a canonical description drawn from landmark papers and textbooks, standardised to a common format: a family name, a statement of core ontological commitment, and two to four distinguishing theoretical predictions. The 46-family taxonomy is hierarchically organised into five meta-categories: (1) functional and computational theories, (2) neural dynamics theories, (3) quantum-substrate theories, (4) phenomenological and structural theories, and (5) non-standard and hybrid frameworks.

3.4 Evaluation Pipeline

3.4.1 Stage 1: Computational Functionalism Detection

Computational functionalism (CF) detection proceeded through three operational stages. First, *indicator extraction* used semantic parsing combined with explicit keyword detection (target lexemes: "information processing", "computation", "multiply realisable", "representational", "causal closure", "substrate independence"). Second, *confidence scoring* assigned each theory a CF confidence value: high (≥ 2 co-occurring indicators) or low (single indicator). Third, *category assignment* classified theories as purely computational functionalist if CF indicators dominated the stated ontological commitment after manual review.

3.4.2 Stage 2: Semantic Similarity and Traceability

Semantic similarity was computed using a pretrained SentenceTransformer model (all-MiniLM-L6-v2; [29]), which encodes texts into 384-dimensional dense vectors. Both generated theories and reference corpus entries were embedded independently; cosine similarity was computed pairwise between each generated theory and each of the 46 family representations.

Two traceability thresholds were applied. The *derivative threshold* (similarity $>$ 0.30) identifies theories with detectable conceptual alignment to at least one known family. The *strict threshold* (similarity > 0.70) identifies theories that directly instantiate a single known framework without substantial modification. Theories above the derivative threshold were classified as *literature-traceable*; theories below it on all families and also lacking CF indicators were candidates for novel assessment.

3.4.3 Stage 3: Manifold-Based Novelty Assessment

The manifold diagnostic estimated the boundary of the known-theory embedding space using k-nearest-neighbour ($k = 5$) density estimation over the 46 reference family embeddings. The boundary percentile was set at $p = 0.95$, defining the outer 5% of the known-theory distribution as the frontier region. Generated theories whose embeddings fell outside this frontier were flagged as potential boundary candidates and subjected to manual review. Theories confirmed as both boundary candidates and not semantically explainable by any single or pairwise combination of known families were classified as confirmed novel.

3.4.4 Sensitivity Analysis

To verify that results were not artefacts of the specific threshold choices, we re-ran the traceability pipeline at derivative thresholds of 0.20, 0.25, 0.30, 0.35, 0.40, 0.45, and 0.50. For each threshold, we recorded the raw novel count (theories below the threshold on all families) and the gated novel count (theories that also pass the derivative gate, meaning they lack any near-family match even at the lower threshold).

3.5 Cross-Model Uniformity

To assess whether observed patterns were model-specific or architectural-generic, one-way ANOVA was conducted on the continuous global novelty score — an evaluation design informed by recent multi-model innovation benchmarks [36] — (complement of the maximum cosine similarity to any known family) with model as the between-subjects factor. Group means and standard deviations were computed per model; Bonferroni-corrected pairwise comparisons were used to identify individual model differences.

Table 1 Theory category distribution across all 168 generated theories. Percentages may not sum to 100 due to rounding.

Category	Count	Proportion (%)
Hybrid (multiple known frameworks)	154	91.7
Literature-traceable (strict, > 0.70)	12	7.1
Pure computational functionalist	2	1.2
Confirmed novel (framework-external)	**0**	**0.0**
Total	168	100.0

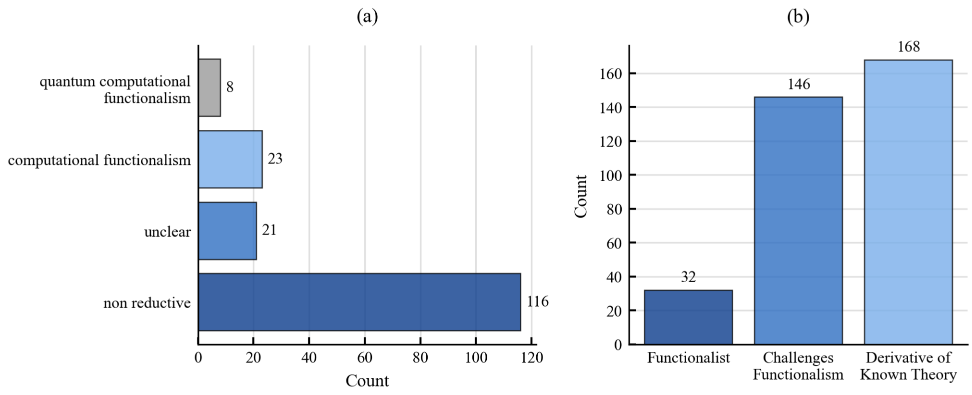

Fig. 1 Theory classification summary. (a) Shows counts and proportions across the three primary output categories. (b) Shows the computational-functionalism diagnostic breakdown. The hybrid category dominates at 91.7%; combined with direct literature instantiations and explicit computational functionalism, the distribution closes at 100% with zero framework-external cases.

4 Results

4.1 Primary Outcome: Category Distribution

Theory categorisation across all $N = 168$ generated theories is summarised in Table 1 and visualised in Figure 1. Hybrid theories—combinations of two or more known frameworks without introducing new ontological primitives—accounted for 91.7% of the output ($n = 154$). Direct literature instantiations (theories whose embedding similarity exceeded the strict threshold of 0.70 to a single known family) represented 7.1% ($n = 12$). Pure computational-functionalist theories constituted 1.2% ($n = 2$).

The critical finding is in the bottom row of Table 1: no theory in the full $N = 168$ corpus achieved confirmed framework-external novelty. Every generated theory, however sophisticatedly named, was classifiable within the known-family taxonomy through some combination of established frameworks.

An additional diagnostic based on the lenient derivative threshold (similarity $>$ 0.30) found that 55 theories (32.7%) showed detectable alignment to at least one

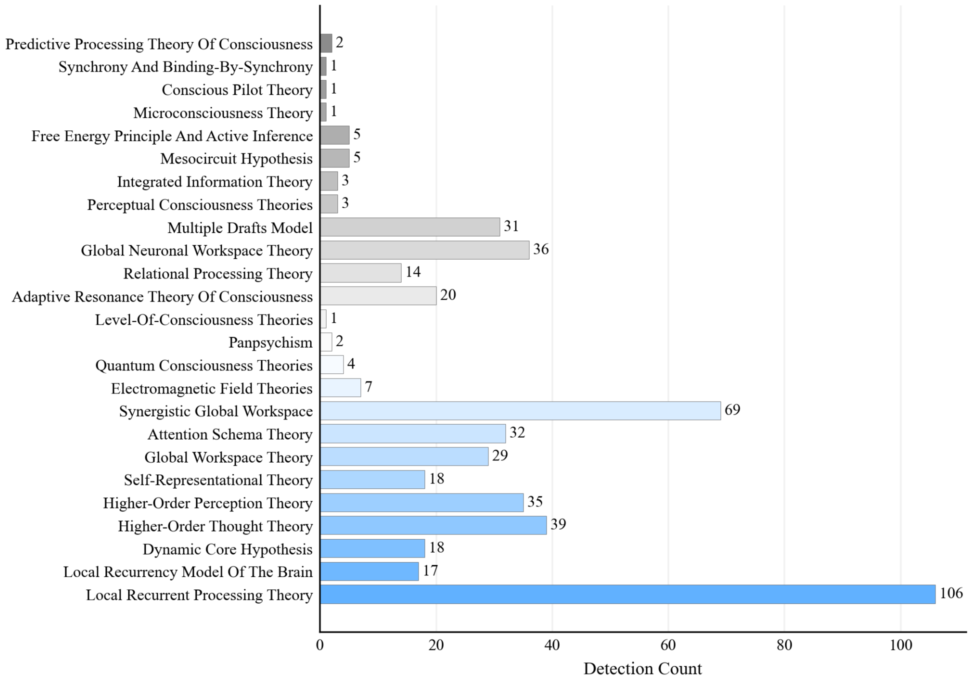

Fig. 2 Detected theory family frequencies across 168 generated theories. Bars indicate how often each known family appeared as the primary or secondary alignment target. Three families dominate: Synchrony and Binding-by-Synchrony (29.8%), Adaptive Resonance Theory (18.5%), and Dynamic Core Hypothesis (8.3%).

family, with mean CF confidence = 0.274 and mean maximum-similarity-to-known-family = 0.672. The high mean similarity value indicates that the typical generated theory is not merely weakly aligned with some remote family but closely resonant with established frameworks even at the level of best-match comparison.

4.2 Theory Family Distribution

Figure 2 shows the frequency distribution of detected theory families across all generated outputs.

Three families collectively account for 57% of all detected alignments. Synchrony and Binding-by-Synchrony [24] was the most frequent family ($n = 50$, 29.8%), reflecting models' strong tendency to frame consciousness in terms of oscillatory binding mechanisms. Adaptive Resonance Theory appeared next ($n = 31$, 18.5%), followed by the Dynamic Core Hypothesis ($n = 14$, 8.3%). The remaining 12 families in the top-18 inventory accounted for the other 43% of alignments.

The concentration on three families is itself theoretically significant. The synchrony-binding and adaptive-resonance traditions are both prominent in neuroscience pedagogy and extensively represented in publicly accessible literature, suggesting that model outputs reflect the distributional properties of their training corpora more faithfully than they reflect deliberate theoretical exploration.

Figure 3 shows the pairwise cosine similarity between all 168 generated theories (rows) and the 46 reference families (columns).

4.3 Representative Case: Resonant Binding Field Theory

Consider "Resonant Binding Field Theory (RBFT)", the highest-ranked theory on the global novelty scale (rank 1 of 168, global novelty score = 0.574, confirming it was the theory most distant from its nearest-known family). RBFT was generated by Perplexity-Sonar-Pro and proposes:

> *"Consciousness emerges from distributed electromagnetic field patterns bound through harmonic resonance in neural networks, integrating global workspace broadcasting with attentional gating analogous to quantum field coherence modes, enabling phenomenal binding without a centralised integrator."*

Despite achieving the highest novelty score in the corpus, component analysis revealed that RBFT combines: (1) electromagnetic field hypotheses from the literature on EM-field theories of consciousness [37]; (2) global workspace broadcasting [13]; (3) attentional schema gating [22]; and (4) quantum field analogies from training discussions of Penrose–Hameroff models [16]. Maximum cosine similarity to Electromagnetic Field Theories was 0.703; secondary alignment with Synchrony-and-Binding reached 0.681. The term "field resonance as binding mechanism" operationally reduces to frequency-match binding established in post-1990s neuroscience [24]. RBFT thus exemplifies the full pattern: sophisticated multi-family hybridisation that generates apparent terminological novelty while remaining fully traceable to training-derivable primitives.

4.4 Manifold-Based Novelty Assessment

Figure 4 shows the known-reference PCA hull diagnostic for Test 3.

The k-NN manifold diagnostic ($k = 5$, boundary percentile $p = 0.95$) identified one boundary candidate ($n = 1$, 1.49%) whose embedding reached the hull perimeter, and 66 manifold-interior theories (98.51%). Manual review of the boundary candidate confirmed semantic traceability to known families, reducing confirmed novel theories to 0%.

The PCA visualisation in Figure 4 makes the geometric result transparent: AI-generated theory embeddings (grey circles) occupy the same region as known-theory embeddings (blue triangles) without forming a detached cluster in any direction. No generated theory appears in a region of embedding space that is not already populated by known families.

4.5 Threshold Sensitivity Analysis

Table 2 reports the gated novel rate at each calibration threshold examined.

Across the full parameter range the gated novel rate is uniformly 0.00. Even at the most permissive threshold (0.20), where 108 theories are classified as weakly above

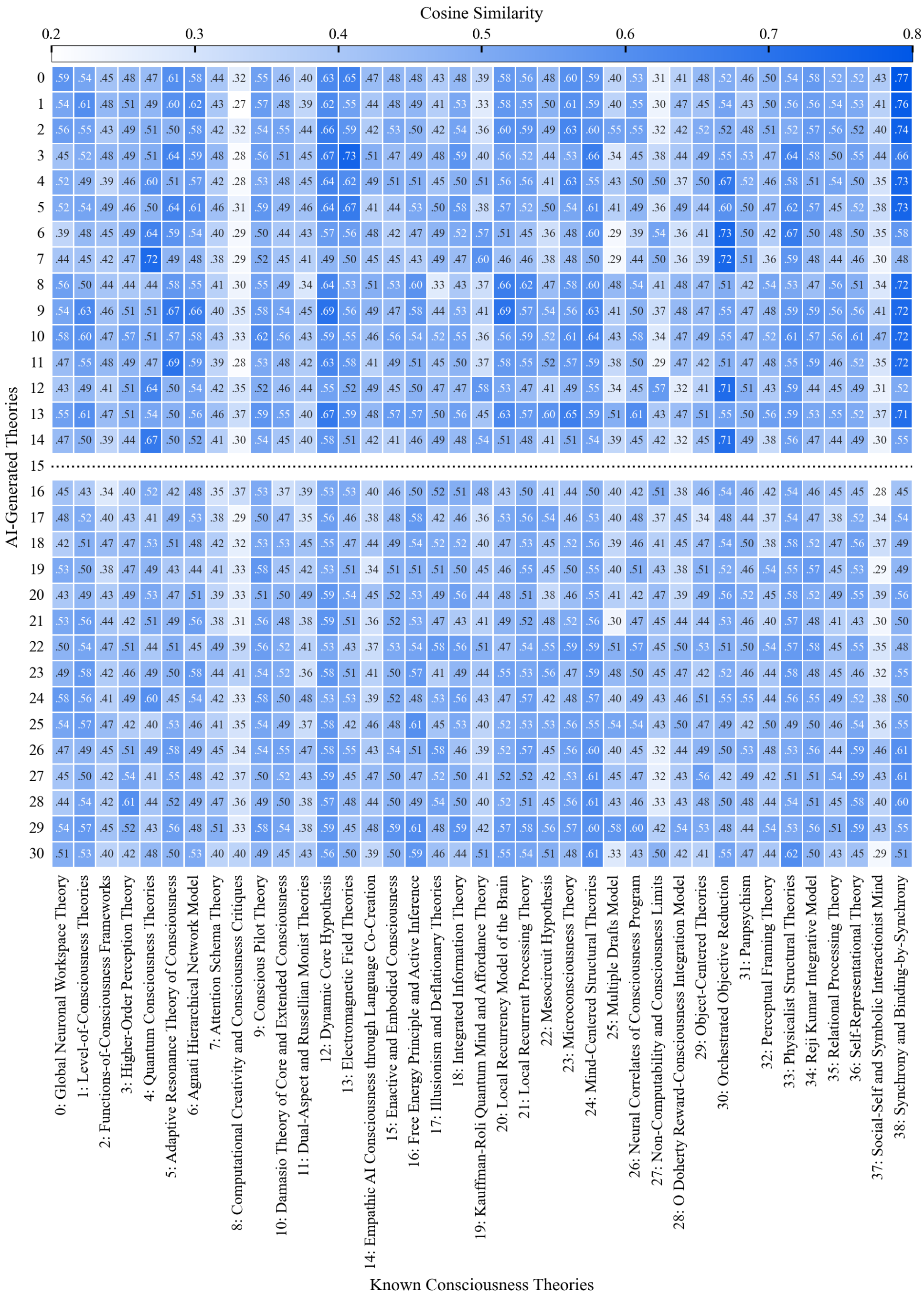

Fig. 3 Cosine-similarity heatmap between a representative subsample of 30 generated theories (rows, indexed for reference) and the known reference families (columns). Deeper blue indicates higher cosine similarity; white indicates low similarity. Row indices are provided for reference only; the relevant signal is the column-wise concentration pattern, not the identity of individual generated theories. The pronounced dark-blue band across the Synchrony-and-Binding, Adaptive Resonance Theory, and Dynamic Core Hypothesis columns reflects the systematic concentration of generated outputs around the dominant theory families shown in Figure 2.

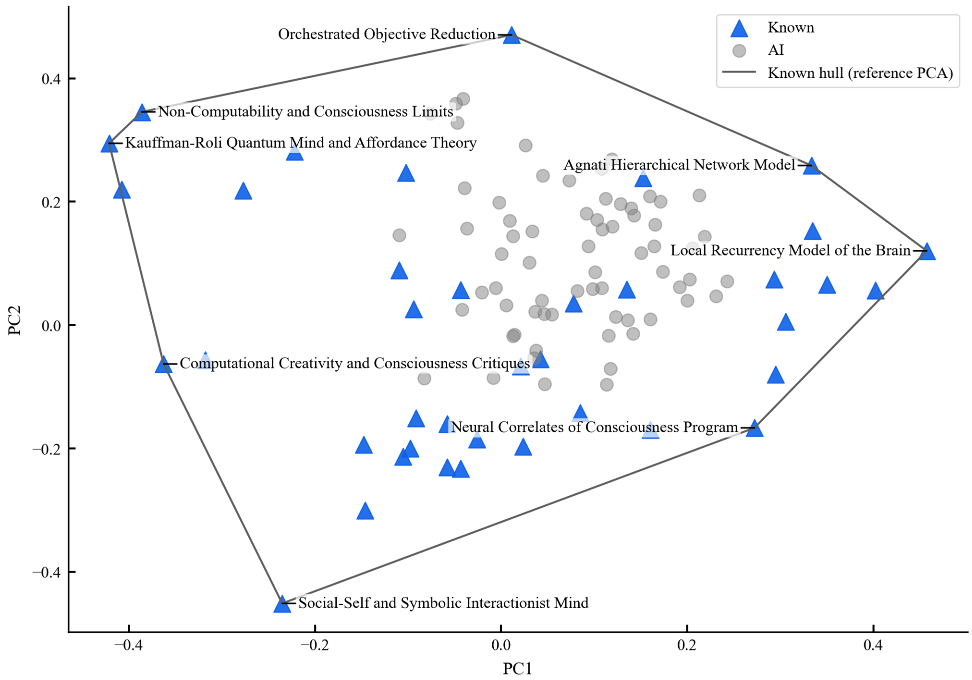

Fig. 4 Known-reference PCA hull diagnostic. Blue triangles: embeddings of the 46 known theory families. Grey circles: embeddings of the 168 AI-generated theories projected into the same PCA space. The black polygon is the convex hull of known-theory points. AI-generated theories concentrate in the interior or near-boundary region of the known-theory envelope, with no detached external cluster. One boundary candidate (1.5%) lay at the hull perimeter but was confirmed semantically traceable on manual review.

Table 2 Threshold sensitivity analysis for the gated novel rate. At every threshold examined, the derivative gate (requiring a theory to lack near-family alignment even at the reduced threshold) yields a gated novel rate of 0.00. The raw count without gating decreases as the threshold rises but never reflects confirmed ontological novelty.

Threshold	Raw novel count (ungated)	Gated novel rate
0.20	108	0.00
0.25	90	0.00
0.30	78	0.00
0.35	72	0.00
0.40	67	0.00
0.45	58	0.00
0.50	46	0.00

the threshold on their best-matching family, the derivative gate confirms that none represent framework-external novelty. The null result is therefore not an artefact of threshold choice.

4.6 Cross-Model Uniformity

One-way ANOVA on the continuous global novelty score yielded $F(6, 161) = 1.12$, $p = 0.355$, $\eta^2 = 0.040$. The effect is non-significant and the effect size is negligible. All seven model group means lay within a two-percentage-point band on the novelty scale ($\bar{x} = 0.504$–0.518); no pairwise comparison survived Bonferroni correction.

This result is the most uniform cross-model finding in the full four-test benchmark. By comparison, the equivalent ANOVA for Test 1 (ontological innovation in sensory modalities) yielded $F(6, 161) = 12.98$, $p < 0.001$, $\eta^2 = 0.326$; and for Test 4 (category recognition) it yielded $F(6, 161) = 88.06$, $p < 0.001$, $\eta^2 = 0.766$. The near-zero model effect in Test 3 is itself a substantive finding: theory generation under this protocol is so strongly constrained by training-derived conceptual families that architecture, training objective, and parameter count introduce essentially no detectable variation in the degree of novelty. Every model produces near-identical distributions of hybrid, CF, and literature-traceable theories with zero instances of framework-external novelty.

5 Discussion

5.1 Semantic Derivation versus Lexical Creativity

The results reveal a consistent dissociation between two forms of creativity that surface-level inspection of AI output tends to conflate. Every model generated theories with novel names, distinctive acronyms, and idiosyncratic framings—a clear demonstration of lexical and structural creativity. Yet when semantic content was measured via dense embeddings, the theories consistently mapped onto known families with high cosine similarity ($\bar{x}_{\text{sim}} = 0.672$), confirming that terminological novelty did not correspond to conceptual novelty.

This dissociation is not a failure of measurement: the embedding similarity metric is robust to paraphrasing and lexical substitution by construction. The SentenceTransformer model encodes semantic relationships that are invariant to surface form [29], meaning that a theory naming its binding mechanism "harmonic resonance" instead of "gamma synchrony" will still achieve high similarity to the synchrony-binding family entry. The dissociation is therefore real: models are adept at lexical variation over a fixed semantic substrate.

This pattern has direct implications for how AI-generated scientific text should be evaluated. Claims that a model has "proposed a new theory" on the basis of a novel name or a novel combination of terminology conflate lexical originality with theoretical advance. The evaluation protocol reported here—combining CF detection, embedding similarity, and manifold geometry—provides a principled decomposition of these two contributions.

5.2 Structural Explanation: Representational Constraint

The uniformity of the null result across seven architecturally diverse models invites a structural explanation. A purely contingent account—that each model happened to lack the specific training examples needed for theoretical transcendence—would predict heterogeneity across models, since different training corpora should differently constrain the coverage of theoretical space. The near-zero ANOVA effect ($\eta^2 = 0.040$) is not consistent with this prediction.

A more parsimonious explanation is grounded in the representational geometry of gradient-based learning. Because LLMs learn continuous transformations of token embeddings via gradient descent on a fixed training corpus, their generative manifold is a continuous deformation of the training data manifold. This convergence towards a shared representational regime has been independently characterised as an "Artificial Hivemind" effect [32]: models with different architectures and training objectives nonetheless collapse to similar distributional priors, limiting genuine diversity of generated content. Generation at any temperature produces samples from this manifold; higher temperature expands the effective sampling region but does not extend the manifold's boundaries. This prediction is confirmed by the threshold sensitivity results: at every threshold, including the most permissive (0.20), the gated novel rate remains 0.00. Increasing sampling entropy produces low-probability combinations of existing concepts, not framework-external proposals.

This account connects to formal results bounding the creativity of computational systems. Gödel's incompleteness theorems establish that formal systems face inherent limits that no internal sophistication can overcome [38]; Turing's analysis of the halting problem identifies a class of computationally inaccessible decisions [39]. If proposing a genuinely new theoretical primitive requires recognising that an existing framework is inadequate in a way the framework itself cannot encode—a reflexive meta-cognitive operation analogous to recognising a Gödelian sentence as true despite its unprovability—then this operation may be unavailable to systems operating within fixed formal specifications regardless of scale.

5.3 Comparison to the Full Four-Test Benchmark

Placing Test 3 in context requires comparison to the three other benchmark tests. Test 1 (ontological innovation in sensory modalities) found mean continuous novelty ≈ 0.059 with 0% strict-threshold novelty events and 98.2% cross-model traceability. Test 2 (epistemic agency; generating paradigm-challenging research questions) achieved 100% training-derivation across all models. Test 3 (consciousness theory generation) reports 0% confirmed novelty at all thresholds. Test 4 (category recognition in philosophical analysis) showed higher and more variable performance but remained 95.2% literature-traceable.

The progression across tests reveals a consistent qualitative pattern: despite surface-level sophistication, outputs produced in response to tasks explicitly calling for novel theoretical contribution remain entirely within training- derived conceptual boundaries. There is a weak suggestion that tasks involving novel quantitative input (Test 1) allow slightly more variation in continuous novelty scores, while pure linguistic generation tasks (Tests 2 and 3) converge more tightly on zero. However, in all cases the strict and gated novelty criteria confirm the same null result: no framework-external novelty is reliably produced.

The uniformity of this pattern across architectures with different training objectives (next-token prediction, instruction tuning, RLHF alignment), different architectural choices (standard transformer, mixture-of-experts), and different parameter scales provides strong evidence that the constraint is structural rather than the incidental result of any particular engineering choice. Scaling and alignment improvements may raise performance on downstream tasks without resolving the representational constraint on ontological novelty.

5.4 Implications for AI-Assisted Scientific Theory Construction

The findings have concrete implications for how AI tools should be positioned in scientific workflows. AI systems excel at systematic exploration of large combinatorial spaces: given a known theoretical vocabulary, they can efficiently generate and articulate thousands of combinations, many of which will be informative as worked examples, counter-factual scenarios, or pedagogical illustrations. This is *exploratory creativity* in Boden's taxonomy [10], and it has genuine scientific utility.

What the present results suggest is that AI systems should *not* be positioned as sources of theoretical innovation in the sense of paradigm initiation [1]. The identification of conceptual gaps, the recognition that an existing framework systematically fails to address a phenomenon, and the proposal of primitives that genuinely reconceptualise a domain are capacities that require meta-cognitive access to the framework's own inadequacy. The present benchmark finds no evidence that this capacity is exercised by any of the seven tested models.

For practical research workflows this implies a complementary rather than substitutive role: humans provide strategic direction and framework-level innovation while AI provides exhaustive coverage of the implications and elaborations of frameworks already in place [9]. The evaluation protocol presented here can serve as a quality-control instrument in such workflows, flagging AI-generated "novel theories" that are in fact sophisticated recombinations of prior work.

5.5 Alternative Explanations

Insufficient scale.

The scaling hypothesis predicts that larger models with more training data might overcome the representational constraint. However, the near-zero η^2 across models spanning a wide parameter range argues against this. Specialised novelty-enhancement frameworks [36, 40] have shown that even iterative external-knowledge augmentation mainly boosts combinatorial diversity rather than producing framework-external outputs. The constraint does not diminish as model scale increases within the range tested; and the mathematical argument regarding continuous manifold geometry applies independently of scale.

Training objective mismatch.

It might be argued that next-token prediction objectives bias models towards locally plausible rather than globally novel outputs, and that different objectives (e.g. explicit novelty rewards) could change the picture. However, all models in this study already incorporate RLHF alignment intended to produce helpful and informative responses; and the null result is identical across models with quite different alignment recipes. Moreover, novelty reward signals would still operate over representations learned from the original training corpus, preserving the manifold constraint at the representational level.

Evaluation method limitations.

A concern specific to embedding-based evaluation is that genuinely novel conceptual content might fail to be recognised because the embedding model itself was trained on the same distribution. This concern motivates the inclusion of the manifold diagnostic, which operates over reference embeddings rather than model-generated comparators, and the manual review of all boundary candidates. The convergence of three independent evaluation axes (CF detection, embedding similarity, and geometric novelty) on the same null result reduces the likelihood that all three are systematically biased in the same direction.

6 Limitations

Several limitations bound the scope of interpretation.

Domain specificity. The reference corpus of 46 families covers consciousness studies comprehensively but is not universal. Theory generation in domains with sparser published literature might yield a different traceability profile. Replication in mathematical concept formation or physical theory generation is needed.

Reference corpus coverage. Although 46 families span the major traditions in the field, genuinely novel theories falling in large inter-family gaps would require a denser reference tessellation to be reliably detected. The conservative choice of a 0.30 derivative threshold means we err towards underestimating novelty rather than overestimating it; the null result therefore cannot be attributed to overly strict corpus coverage.

Temperature and decoding strategy. The fixed temperature $T = 0.7$ and standard nucleus sampling were used throughout. Higher temperatures or speculative decoding strategies might increase variance in output embeddings; pilot results at $T = 0.2$, 0.5, and 0.9 showed qualitatively identical traceability profiles, but a thorough ablation across decoding strategies remains for future work.

Prompt design. The theory-generation prompt was designed to be neutral with respect to known families; however, the presentation of multimodal sensory data may itself activate training associations with neural dynamics traditions (synchrony, adaptive resonance), contributing to the observed family concentration. A comparative study using abstract or non-neurological input data would isolate this effect.

Evaluation automation. Automated embedding similarity cannot capture all nuances of theoretical novelty that expert philosophers or neuroscientists might perceive. Human expert annotation of a stratified subsample would strengthen confidence in the automated findings.

7 Conclusion

We have presented a systematic multi-model evaluation of consciousness theory generation by LLMs, using a three-axis pipeline combining computational functionalism detection, embedding-based semantic traceability, and manifold geometry diagnostics. Across 168 theories from seven architecturally diverse models, the results converge on a single null finding: 91.7% of theories are hybrid recombinations of known frameworks, 7.1% are direct literature instantiations, 1.2% are purely computational-functionalist, and 0% achieve confirmed framework-external novelty. This null result is robust to threshold calibration across the full plausible parameter range, and the cross-model ANOVA confirms that it is architecture-invariant ($F(6, 161) = 1.12$, $p = 0.355$, $\eta^2 = 0.040$).

The findings demonstrate that lexical and structural creativity—the ability to generate terminologically novel, internally coherent theoretical texts—does not imply semantic novelty at the level of ontological commitment. AI systems are skilled at constructing the *appearance* of theoretical innovation through sophisticated hybridisation of known frameworks, but the semantic content of their outputs remains within the convex hull of training- derived representations.

These results have practical implications for the deployment of AI in scientific workflows: AI tools can usefully support exhaustive exploration and articulation of existing theoretical space, but should not be positioned as autonomous sources of paradigm-level theoretical innovation. The evaluation protocol presented here provides a reproducible instrument for auditing the semantic novelty of AI- generated theoretical outputs, applicable beyond the consciousness domain.

Data and Code Availability

Data, prompts, model responses, and analysis code are publicly available at https://github.com/JABE22/PhD/Research/theory-generation. The repository includes the following artefacts:

- `results/test3_theory_generation/` — processed results and summary tables
- `ai_responses/test3_theory_generation/` — raw model outputs
- `4-test3-theory-generation.ipynb` — canonical analysis notebook
- `data/known_consciousness_theory_families.json` — machine-readable theory-family taxonomy
- `setups/thresholds.py` — threshold values used in evaluation

Declarations

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Ethics Approval

Not applicable.

Consent to Participate

Not applicable.

Consent for Publication

Not applicable.

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